



3.7.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.



Unit 3, Lesson 8: Writing Large Numbers in Scientific Notation



Benchmarks

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.



Learning Targets

- ☐ I can use the laws of exponents to evaluate numerical expressions with a base of 10 and exponents greater than or equal to 0.
- ☐ I can express large numbers using scientific notation.
- ☐ I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.



3.8.1 Warm-Up: Grains of Sand

The beaches on Siesta Key are 99% quartz, meaning that the beaches are made of some of the finest grains of sand.



1. Suppose you have a teaspoon of the sand from Siesta Key. How could the teaspoon of sand be used to estimate the number of grains of sand in a bucket of sand?

A group of researchers at the University of Hawaii used a teaspoon of sand to estimate the number of grains of sand on all of the beaches on the Earth. Their estimate was that there are roughly seven quintillion, five hundred quadrillion grains of sand. A quintillion is 1,000,000,000,000,000,000 and a quadrillion is 1,000,000,000,000,000.

2. Write seven quintillion, five hundred quadrillion.



3.8.2 Exploration: Powers of Ten

1. Use a scientific calculator to find the value of each power of 10.

Power of 10	Value
10^1	
10^2	
10^3	
10^4	
10^5	
10^6	
10^7	
10^8	
10^9	
10^{10}	

2. What do you notice about the relationship between the power of 10 and its value?

3. Work with your partner to complete the table.

Power of 10	Value	Place Value of 1
10^1	10	Tens
10^2		
10^3		
10^4		
10^5		
10^6		

4. Work with your partner to describe how the exponent of the power of ten relates to place value.

5. Based on the pattern in the table, what is the value of 10^0 ?



3.8.3 Exploration: Patterns in Scientific Notation

Very large numbers can be written using powers of 10 in a shorthand method called scientific notation.

1. Work with your partner to complete the table.

Power of 10	Value	Scientific Notation	Word Form
10^2	100	1.0×10^2	one hundred
10^3		1.0×10^3	
		$1.0 \times 10^{\boxed{}}$	ten thousand
	100,000	$\boxed{} \times 10^5$	
10^6		$\boxed{} \times 10^{\boxed{}}$	
	1,000,000,000		
		1.0×10^{12}	one trillion

2. Select two of the numbers. Work with your partner to name something in the world that is described by those numbers.

Number	What can be described by this number in the real world?

3. Work with your partner to describe at least two patterns in the relationships among the power of 10, the numerical value, the scientific notation, and/or the word form.

a.

b.



3.8.4 Guided Instruction: Writing Large Numbers in Scientific Notation

1. Consider the number 2,000,000.

a. What power of ten represents 1 million? $10^{\boxed{}}$

- b. Complete the multiplication sentence to express 2,000,000 using this power of 10.

$$\boxed{} \times 10^{\boxed{}} = 2,000,000$$



Scientific notation is a method of writing very large or very small numbers using exponents in which a number is expressed as the product of a power of 10 and a number that is at least 1 and less than 10.

2. Complete the statement.

When rewriting 2,000,000 in scientific notation, the factor

2 is ☐ less than
☐ greater than or equal to 1 and

☐ less than
☐ greater than or equal to 10. The digit 2 is in the

millions place. A million expressed as a power of 10 is

☐ 10^5 .
☐ 10^6 .

Therefore, 2,000,000 is written as ☐ 2×10^5
☐ 2×10^6

in scientific notation.

3. The expression 20×10^5 is also equivalent to 2,000,000. Use the definition of scientific notation to explain why 20×10^5 is not the correct expression of 2,000,000 in scientific notation.

4. Consider the standard form and scientific notation of the numbers shown in the table.

Standard Form	Scientific Notation
2,300,000	2.3×10^6
2,030,000	2.03×10^6

Describe the similarities and differences between the values expressed in scientific notation.

Similarities	Differences

As explored in the previous lesson, the powers of 10 relate directly to place values.

5. The planet Venus is about 26,000,000 miles from earth.
- What value between 1 and 10 could be multiplied by a power of 10 to result in 26,000,000?
 - Which digit is in the ones place of your answer to Part A?
 - In which place value is that digit in 26,000,000?
 - Which power of 10 relates to this place value?
 - Express 26,000,000 in scientific notation.
6. In 2018, funding for education in Florida totaled \$30,100,000,000. Express this value in scientific notation.



3.8.5 Collaboration: Comparing Large Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

a. Complete the table.

Scientific Notation	Standard Form
	200
2×10^3	
2×10^4	
	200,000

b. Complete each statement.

2×10^3 is
☐ 10
☐ 100
☐ 1000
 times
☐ smaller
☐ larger
 than 2×10^2 .

2×10^2 is
☐ 10
☐ 100
☐ 1000
 times
☐ smaller
☐ larger
 than 2×10^4 .

2×10^5 is
☐ 10
☐ 100
☐ 1000
 times
☐ smaller
☐ larger
 than 2×10^2 .

c. Describe how to compare numbers written in scientific notation if the factors being multiplied by a power of 10 are the same.

2. Complete the table.

Scientific Notation	Standard Form
	32,000,000
6.4×10^7	
	96,000,000

a. How many times larger is 6.4×10^7 than 3.2×10^7 ?

b. How many times smaller is 3.2×10^7 than 9.6×10^7 ?

c. Describe how to compare numbers written in scientific notation if the powers of 10 are the same.

3. Complete the table.

Scientific Notation	Standard Form
2.1×10^4	
	420,000
8.4×10^6	

a. How many times larger is 4.2 than 2.1?

b. How many times larger is 10^5 than 10^4 ?

c. How many times larger is 4.2×10^5 than 2.1×10^4 ?

 $\times$ = _____ times larger

d. Complete the statement.

The value of 8.4×10^6 is

- ☐ 4
- ☐ 40
- ☐ 400

times larger than

the value of 2.1×10^4 because 8.4 is

- ☐ 4
- ☐ 40
- ☐ 400

times

larger than 2.1, 10^6 is

- ☐ 10
- ☐ 10^2
- ☐ 10^3

times larger than

10^4 , and $\times$ =

- ☐ 4.
- ☐ 40.
- ☐ 400.

If the factors being multiplied by the powers of 10 are different, but one is not a multiple of the other, the comparison of the two values can be *approximated*.

4. The average distance from Earth to the moon is 2.38855×10^5 miles. The average distance from Earth to the sun is about 9.3×10^7 .

Complete the statements to approximate how many times farther the sun is from Earth than the moon.

9.3 is about

- ☐ 2
- ☐ 4
- ☐ 6

times

- ☐ smaller
- ☐ larger

than 2.38855 and

10^7 is

- ☐ 10
- ☐ 10^2
- ☐ 10^3

times

- ☐ smaller
- ☐ larger

than 10^5 , so the value

of 9.3×10^7 is about _____ times larger than 2.38855×10^5 .



3.8.6 Wrap-Up: Cool Down

1. Explain how to express large numbers using scientific notation in your own words.